

3.1.3 Bonding

The physical and chemical properties of compounds depend on the ways in which the compounds are held together by chemical bonds and by intermolecular forces. Theories of bonding explain how atoms or ions are held together in these structures. Materials scientists use knowledge of structure and bonding to engineer new materials with desirable properties. These new materials may offer new applications in a range of different modern technologies.

3.1.3.1 Ionic bonding

Content	Opportunities for skills development
Ionic bonding involves electrostatic attraction between oppositely charged ions in a lattice. The formulas of compound ions, eg sulfate, hydroxide, nitrate, carbonate and ammonium. Students should be able to: • predict the charge on a simple ion using the position of the element in the Periodic Table • construct formulas for ionic compounds.	

3.1.3.2 Nature of covalent and dative covalent bonds

Content	Opportunities for skills development
A single covalent bond contains a shared pair of electrons. Multiple bonds contain multiple pairs of electrons. A co-ordinate (dative covalent) bond contains a shared pair of electrons with both electrons supplied by one atom. Students should be able to represent: • a covalent bond using a line • a co-ordinate bond using an arrow.	

3.1.3.3 Metallic bonding

Content	Opportunities for skills development
Metallic bonding involves attraction between delocalised electrons and positive ions arranged in a lattice.	

3.1.3.4 Bonding and physical properties

Content	Opportunities for skills development
The four types of crystal structure: • ionic • metallic • macromolecular (giant covalent) • molecular. The structures of the following crystals as examples of these four types of crystal structure: • diamond • graphite • ice • iodine • magnesium • sodium chloride. Students should be able to: • relate the melting point and conductivity of materials to the type of structure and the bonding present • explain the energy changes associated with changes of state • draw diagrams to represent these structures involving specified numbers of particles.	AT a, b, h and k PS 1.1 Students could be asked to find the type of structure of unknowns by experiment (eg to test solubility, conductivity and ease of melting).

3.1.3.5 Shapes of simple molecules and ions

Content	Opportunities for skills development
Bonding pairs and lone (non-bonding) pairs of electrons as charge clouds that repel each other. Pairs of electrons in the outer shell of atoms arrange themselves as far apart as possible to minimise repulsion. Lone pair–lone pair repulsion is greater than lone pair–bond pair repulsion, which is greater than bond pair–bond pair repulsion. The effect of electron pair repulsion on bond angles. Students should be able to: • explain the shapes of, and bond angles in, simple molecules and ions with up to six electron pairs (including lone pairs of electrons) surrounding the central atom.	MS 0.3 and 4.1 Students could be given familiar and unfamiliar examples of species and asked to deduce the shape according to valence shell electron pair repulsion (VSEPR) principles.

3.1.3.6 Bond polarity

Content	Opportunities for skills development
Electronegativity as the power of an atom to attract the pair of electrons in a covalent bond. The electron distribution in a covalent bond between elements with different electronegativities will be unsymmetrical. This produces a polar covalent bond, and may cause a molecule to have a permanent dipole. Students should be able to: • use partial charges to show that a bond is polar • explain why some molecules with polar bonds do not have a permanent dipole.	

3.1.3.7 Forces between molecules

Content	Opportunities for skills development
Forces between molecules: • permanent dipole–dipole forces • induced dipole–dipole (van der Waals, dispersion, London) forces • hydrogen bonding. The melting and boiling points of molecular substances are influenced by the strength of these intermolecular forces. The importance of hydrogen bonding in the low density of ice and the anomalous boiling points of compounds. Students should be able to: • explain the existence of these forces between familiar and unfamiliar molecules • explain how melting and boiling points are influenced by these intermolecular forces.	AT d and k PS 1.2 Students could try to deflect jets of various liquids from burettes to investigate the presence of different types and relative size of intermolecular forces.